



Sarvajanik University

Sarvajanik College of Engineering and Technology



Department of Information Technology / Artificial Intelligence and Data Science

Summer Project



Real-Time AI-Powered Meeting Intelligence Platform

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Introduction

- Effective communication intelligence is critical in modern professional environments.
- AI-driven tools can unlock real-time insights from live conversations.
- TalkSense AI bridges the gap between raw speech and actionable meeting intelligence.
- Built as a full-stack, production-grade system using modern AI and web technologies.

Objective:

- Design and implement a real-time, AI-powered platform that transcribes, analyzes, and provides live insights from ongoing conversations.

Problem Definition

- Current tools analyze conversations only **after** meetings end — insights arrive too late.
- No live coaching, speaker feedback, or engagement tracking during conversations.
- Manual evaluation remains subjective, inconsistent, and unscalable.
- Recruiters, sales teams, and managers need **real-time conversation intelligence**.

Current State	Required State
Post-meeting analysis	Real-time analysis
Delayed insights	Live insights
Manual evaluation	Automated evaluation
No live feedback	Live coaching

Motivation

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- Sales teams lose deals due to lack of real-time conversation guidance.
 - Recruiters need objective, data-driven interview assessments.
 - Remote teams struggle with engagement and collaboration tracking.
 - AI advancements (Whisper, Transformers) make real-time speech intelligence feasible today.
 - A modular, scalable platform could serve hiring, sales, coaching, and team collaboration.

Proposed Solution & Key Features

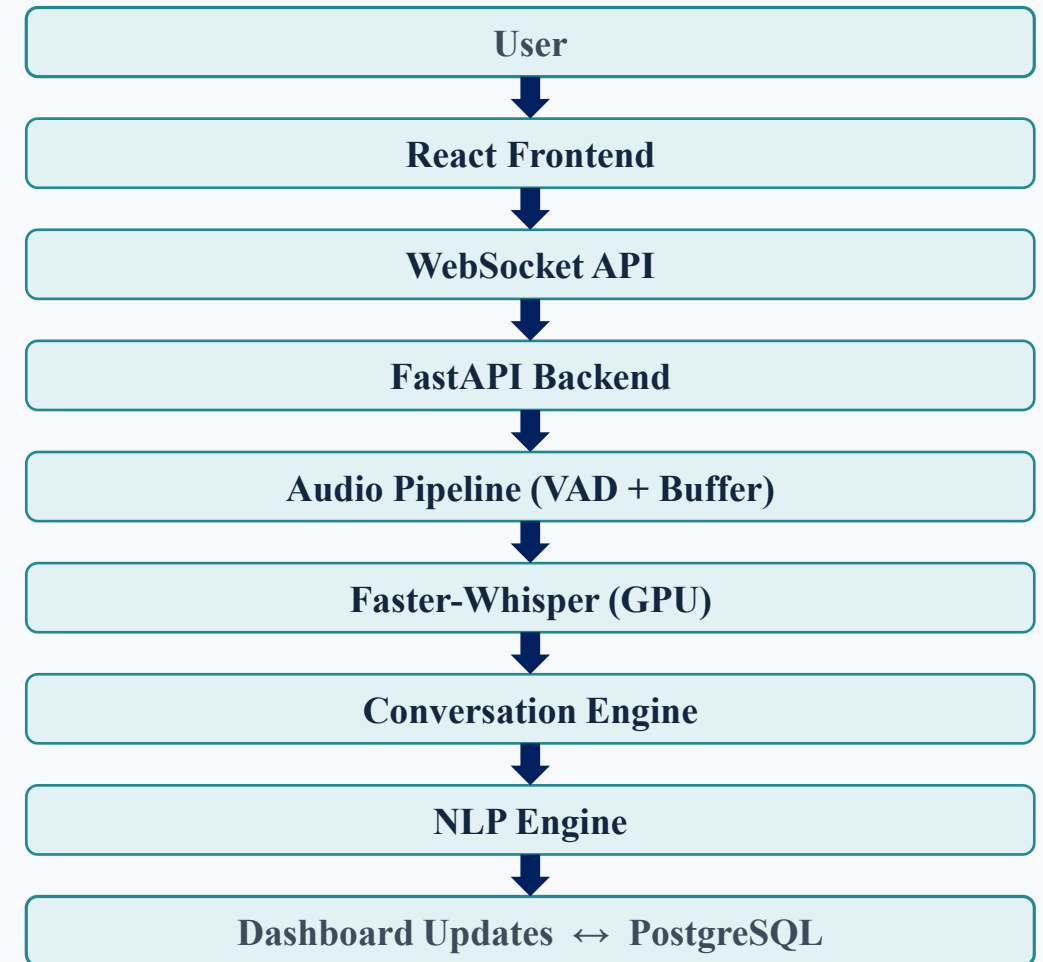
- **Live speech transcription → instant conversation understanding**
- **NLP engine: sentiment, filler words & speaker analytics — continuously**
- **Actionable insights surfaced during live conversations**
- **Modular: supports streaming, storage & live dashboards**

❖ Key Features

- | | | | |
|-----------------|-----------------|--------------------|---------------------|
| • Transcription | • Sentiment | • Filler Detection | • Speaker Analytics |
| • Sales Intel | • Meeting Intel | • Live Coaching | • Live Dashboard |

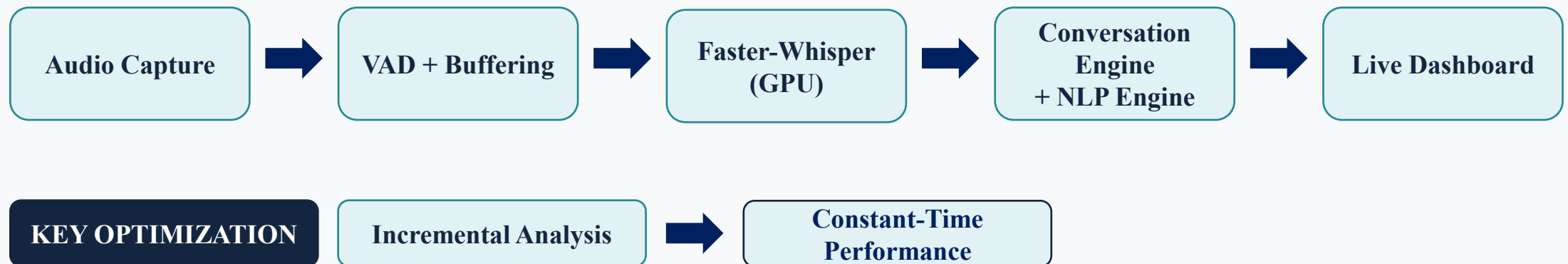
System Architecture

- **Modular layered design** separates audio, AI, analytics, and storage concerns.
- **Low-latency WebSocket streaming** connects the frontend and backend in real time.
- **GPU-accelerated Faster-Whisper** delivers word-level speech transcription.
- **Independent NLP and Conversation Engines** enable extensible analytics processing.



AI Pipeline & Workflow

- **Background VAD + Buffering** keeps audio streaming responsive and unblocked.
- **Optimized GPU scheduling** enables fast, parallel speech transcription.
- **Intelligent transcript merging** eliminates duplicate speech at processing boundaries.
- **Incremental analysis** processes only new data — constant-time at any meeting length.



Technology Stack



Layer	Technology	How We Used It
Frontend	React + TypeScript + Vite	Built live dashboard with real-time WebSocket updates
Backend	FastAPI + Python	Designed async REST + WebSocket APIs for streaming
Database	PostgreSQL + SQLAlchemy	Modeled session, transcript & analytics data
AI / ML	Faster-Whisper + Hugging Face Models	GPU-accelerated transcription + NLP sentiment analysis
CI/CD	GitHub Actions	Automated code validation on every commit

Implementation & Results

TalkSense AI Home Analyze History

"User configures mode & audio source before session"

Launch Live Intelligence Session

Configure your workspace and review client memory before starting the live session.

Conversation Mode

Meeting

Sales Call

Interview

Client Briefing & Memory

Meeting Mode selected

Internal team meeting mode does not use client relationship briefing memory.

Audio Source

☒ Microphone

Capture audio from the default microphone.

☐ System Audio

Capture audio from a browser tab or the system.

☐ Microphone + System Audio (Meeting Mode)

Capture local microphone and remote meeting participants simultaneously.

Launch Live Session

TalkSense AI Home Analyze History Interview Completed Idle Ending...

Session Recording

0:00 / 0:39

Live Transcript

Thank you for watching this video.

Participant: [7.7s - 8.6s] Positive
Thank you for watching.

Participant: [11.5s - 12.7s] Neutral
Thank you.

Participant: [14.3s - 14.6s] Neutral
Thank

Host: [18.0s - 19.5s] Positive
Hello, I know you can hear me,

Host: [20.0s - 20.7s] Neutral
please answer.

Participant: [22.0s - 22.3s] Neutral
Hello.

Host: [24.8s - 25.0s] Neutral

Live Coaching

Speaking Balance
Pause and ask an open question.
You are speaking 100% of the time.

Real-Time Alerts
No active alerts.

Session Metrics

SESSION HEALTH 100%

Sentiment: 62% Filler Penalty: 0
Pause Penalty: 0 Action Items: 0

Speaking Ratio
Balanced: Speaking is well balanced.

Host 47.4%
Participant 52.6%

Conversation Flow (Live)

Switches: 6 Interruptions: 0
Fillers: 0 Duration: 48.4s

Live

"Real-time transcription, coaching & metrics during live session"

Implementation & Results

- **Speech-to-Text (Faster-Whisper)**

- Time-to-First-Word Latency: ~3.5 seconds
- Sub-second word-level timestamp precision
- Complexity: $O(1)$ constant-time performance

- **Speaker Attribution (Pyannote 3.1)**

- Overall Accuracy: 84.0%
- Transcript Coverage: 84.0%
- Macro F1-Score: 0.781

- **NLP & Conversation Engine**

- Live Sentiment Analysis: Accurate per-utterance (updates Session Health instantly)
- Role Classification: LLM-based experimental feature (~25% baseline)
- Objection Handling: 4-tier response scoring implemented

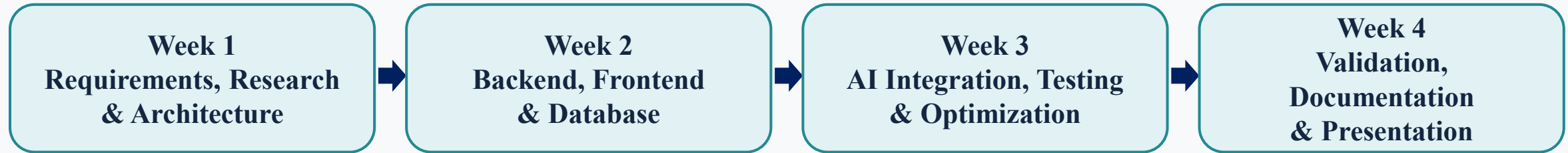
- **Infrastructure & Pipeline Health**

- Backend Cold Start: ~10s (loads Whisper, Pyannote, Sentiment models)
- WebSocket Init Latency: < 1 second
- Live Audio Sync: 100% accurate alignment via PCM audio rewriting
- CI/CD: Automated validation pipeline active

Project Timeline & Future Scope



❖ PROJECT TIMELINE



FUTURE SCOPE

- Live speaker identification & diarization

- Cloud deployment

- Multi-language support

- AI meeting summaries

- Mobile application

- Organization analytics

References



1. FastAPI Documentation — <https://fastapi.tiangolo.com>
2. React Documentation — <https://react.dev>
3. PostgreSQL Documentation — <https://www.postgresql.org>
4. SQLAlchemy — <https://www.sqlalchemy.org>
5. Faster-Whisper — <https://github.com/SYSTRAN/faster-whisper>
6. Hugging Face Transformers — <https://huggingface.co/docs/transformers>
7. GitHub Actions — <https://docs.github.com/en/actions>